

Assessing Bicycle Lane Quality Using Smartphone Sensors

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1. Business Understanding and Objective

Cycling brings physical activity into everyday travel. Regular cycling is associated with lower overall mortality and a lower risk of cardiovascular death (Zhao et al., 2021). Infrastructure quality can influence the experience of cycling. Surface condition has been linked to people's willingness to cycle (Niska et al., 2024). Uneven road surfaces increase the vibrations experienced by cyclists and reduce riding comfort (Niska et al., 2024; Ahmed et al., 2024). Reliable information on the condition of bicycle lanes is needed to determine where maintenance is required and to prioritise improvements (Niska et al., 2024). Pavement condition deteriorates over time due to ageing, traffic loading and environmental conditions and therefore needs to be monitored regularly (Al-Sabaei et al., 2024). However, manual pavement inspection is time-consuming and labour-intensive (Ahmed et al., 2024). Smartphones offer a way to collect pavement-condition data while cycling since their built-in accelerometers can capture vibrations caused by pavement roughness, allowing bicycle infrastructure to be assessed at lower cost and on a larger scale (Ahmed et al., 2024). The aim of this project is to use smartphone sensor data and machine-learning models to classify road sections used by cyclists as smooth or bumpy. This automated assessment could help identify poor-quality sections and support more efficient maintenance of cycling infrastructure.

2. Data Understanding

2.1 Dataset characteristics and collection methodology

Data were collected by three participants, each using their own bicycle and iOS smartphone. Before data collection, the participants met on campus to test the measurement setup and agree on a consistent collection procedure. The Sensor Logger application was used to record accelerometer, gyroscope and gravity data at a sampling rate of 100 Hz. The standardisation option was enabled and all other sensors were turned off. The phone was placed in the right rear trouser pocket for all participants, with the same orientation. This position was chosen after testing a jacket pocket, where the phone moved more during cycling.

The participants collected data independently on different roads rather than repeatedly riding the same routes. This was done to include a wider variety of road conditions and better reflect the different surfaces cyclists encounter in daily use. These included dedicated bicycle lanes and ordinary roads commonly used by cyclists. Smooth sections had relatively even surfaces, while bumpy sections included uneven asphalt, cracks, potholes and uneven paving stones. The selected bumpy sections represented road conditions that cyclists may encounter during regular cycling and were not considered to be unsafe to ride on. Each recording was labelled as either smooth or bumpy based on the overall surface condition of the road section. Participants cycled at their normal speed so that the measurements reflected regular cycling behaviour. Recording duration depended on the length of the road and therefore varied between recordings. The raw dataset contained 30 recordings and 420,506 measurements. In total, 20 recordings were labelled as bumpy and 10 as smooth. The bumpy recordings had a total duration of 45.3 minutes. The smooth recordings covered 24.8 minutes in total.

2.2 Exploratory data analysis and key insights

Acceleration magnitude was calculated from the three accelerometer axes to compare the signal patterns of smooth and bumpy recordings. In the example recordings, the bumpy road showed more frequent and larger fluctuations in acceleration. The smooth recording still contained some peaks, but these were generally smaller and the overall variation was lower. This indicates that individual peaks can also occur on smooth roads and do not necessarily indicate a bumpy surface. Large changes in acceleration were also visible at the beginning and end of the recordings due to placing the phone in and removing it from the pocket, rather than changes in road condition.

2.3 Data quality assessment and limitations

Checks of the raw sensor files found no missing values or duplicate timestamps, and all timestamps were in increasing order. Estimated sampling rates were close to 100 Hz (99.75–100.14 Hz). Some limitations remain in the collected data. Each recording was given one smooth or bumpy label based on the overall road condition, so short sections within a recording may not fully match the assigned label, especially after the recordings are divided into shorter windows. The phone position and orientation were kept consistent for all participants, but the phone was not rigidly fixed and could still move during cycling, which may have influenced the sensor measurements. The dataset was also imbalanced, with 20 bumpy and 10 smooth recordings and more total recording time for the bumpy class. Data were collected by only three participants, which limits the range of cyclists represented in the dataset. Participants cycled at their normal speed, but speed was not recorded, so some variation in the measurements may also be related to cycling speed rather than road condition.

3. Data Preparation

3.1 Preprocessing pipeline and rationale

The accelerometer, gyroscope and gravity data from each recording were combined after checking that their timestamps matched. The first 4 seconds and last 6 seconds of each recording were then removed to exclude the periods when the phone was placed in and removed from the pocket. A longer interval was removed at the end because taking the phone out and stopping the recording took longer than placing the phone in the pocket at the start. After trimming, 390,501 of the original 420,506 measurements remained across the 30 recordings.

3.2 Feature engineering/extraction/selection strategy

The recordings were divided into non-overlapping 2-second windows. A 2-second window was chosen to keep each window focused on a short part of the ride while still containing enough measurements to describe the sensor behaviour. Windows with fewer than 180 measurements were removed so that feature extraction was based on windows of similar length. After filtering, 1,941 windows remained. Acceleration and gyroscope magnitudes were calculated from the x, y and z axes to represent the overall movement recorded by each sensor. For each 2-second window, the mean and standard deviation were calculated for the accelerometer, gyroscope and gravity signals. The same statistics were also calculated for the accelerometer and gyroscope magnitudes. In addition, the maximum and root mean square (RMS) values were calculated for both magnitudes. This resulted in 26 numerical features for each window. Mutual information was used to score each feature based on how much knowing that feature reduced uncertainty about the road-condition label. For supervised

modelling, the 10 highest-scoring features were selected. To avoid data leakage, feature selection was fitted using only the training data.

4. Modelling

4.1 Supervised learning approach and tuning

Four supervised models were used to classify the road sections as smooth or bumpy: K-Nearest Neighbours (KNN), Decision Tree, Random Forest and Logistic Regression. KNN classifies a window based on nearby observations in the feature space and was included to see whether windows with similar sensor measurements also had the same road condition. Decision Tree learns decision rules from the input features and was chosen because it can model nonlinear relationships while keeping the resulting model interpretable. Random Forest combines multiple decision trees and was included to test whether combining several trees could improve stability compared with a single tree. Logistic Regression estimates the probability of the road condition based on a weighted combination of the input features and was included as a linear baseline for comparison.

Mutual-information feature selection was performed within the training data during model fitting to prevent the validation and test data from influencing which features were selected. KNN and Logistic Regression were standardised using *StandardScaler* because both models are sensitive to differences in feature scales. Decision Tree and Random Forest were trained on the unscaled selected features. For each model, GridSearchCV tested the different hyperparameter combinations shown in Table 1. GroupKFold was used during this search to keep all windows from the same participant together, so the data from one participant was not split between the training and validation sets. The hyperparameter combination with the highest balanced accuracy was selected. Balanced accuracy was used because the dataset contained more bumpy than smooth windows.

Table 1: Hyperparameters tested for the supervised models.

Model	Hyperparameters tested
KNN	n_neighbors: 3, 5, 7, 9, 11, 15, 21
Decision Tree	max_depth: 3, 5, 7, 10, None; min_samples_leaf: 1, 5, 10, 20; class_weight: None, balanced
Random Forest	n_estimators: 100, 200; max_depth: 5, 10, None; min_samples_leaf: 1, 5, 10; class_weight: None, balanced
Logistic Regression	C: 0.01, 0.1, 1, 10, 100; class_weight: None, balanced

4.2 Unsupervised learning approach and parameter selection

Four unsupervised models were used to identify patterns in the data: Self-Organising Map (SOM), DBSCAN, K-Means and Gaussian Mixture Model (GMM). SOM maps similar windows to nearby nodes on a two-dimensional grid and was included to examine whether similar sensor patterns formed clear groups in the data. DBSCAN groups windows that occur in dense regions of the feature space and can identify windows that do not belong to a cluster as noise. It was included because it does not need the number of clusters to be specified in advance and it can identify observations that do not clearly belong to a cluster. K-Means groups windows based on their distance to the cluster centres and

was included as a simple distance-based clustering method. GMM groups windows based on the probability that they belong to each component and was included to compare this probabilistic approach with distance-based clustering.

The unsupervised models used all 26 extracted features. Mutual-information feature selection was not used because it depends on the known road-condition labels. The features were standardised using *StandardScaler* before clustering so that differences in feature scale did not dominate the clustering results. SOM was trained using fixed settings: a 6×6 grid with a sigma of 1.5, a learning rate of 0.5 and 10,000 iterations. K-Means and GMM were both set to two groups because the dataset contained two road conditions, but the models were not given any information about which group represented smooth or bumpy roads. K-Means used 20 initialisations and a maximum of 500 iterations. GMM used 10 initialisations, a diagonal covariance structure and a maximum of 500 iterations. Different combinations of *eps* and *min_samples* were tested for DBSCAN. The highest silhouette score of 0.382 was obtained with *eps* = 1.5 and *min_samples* = 20, but 66.3% of the windows were classified as noise. Using *eps* = 3.0 and *min_samples* = 5 gave a similar silhouette score of 0.371 while reducing the amount of noise to 9.7%. These settings were selected as a better balance between cluster separation and retaining more of the data.

4.3 Comparison methodology

The supervised models were compared based on their ability to classify smooth and bumpy road conditions on unseen participants. The unsupervised models were evaluated separately based on cluster separation and how well the resulting clusters matched the known road-condition labels. Because supervised and unsupervised metrics measure different outcomes, they were not combined into a single model ranking. The two approaches were instead compared based on their suitability for the road-condition classification task. Classification performance was the main criterion for selecting the final model, since the deployed model had to predict whether a road section was smooth or bumpy. Interpretability and computational efficiency were also considered when assessing practical suitability.

5. Evaluation

5.1 Performance metrics and validation strategy

Leave-One-Subject-Out (LOSO) cross-validation was used to evaluate the supervised models. One participant was held out for testing in each fold and the remaining two participants were used for training, feature selection and hyperparameter tuning. Each participant was used as the test participant once. This reduces the influence of participant-specific patterns on the evaluation. Model performance for each test participant was measured using accuracy, balanced accuracy, precision, recall and F1-score. Bumpy was treated as the positive class. Balanced accuracy was used as the main metric because of the class imbalance in the dataset. The mean and standard deviation across the three participants were used to summarise the overall performance of each model.

The unsupervised models were evaluated using silhouette score, Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI). Silhouette score was used to assess how well the clusters were separated. ARI and NMI were used to measure how well the resulting labels matched the known road-condition labels. These labels were only used for evaluation after clustering. Participant ARI was

also calculated to check whether the clusters were more related to participant differences than to road condition. For DBSCAN, the percentage of windows classified as noise was also included because these windows were excluded when calculating the clustering metrics.

5.2 Model comparison and final model selection

Overall supervised model performance is reported in Table 2. Decision Tree achieved the highest mean balanced accuracy (0.743 ± 0.168), followed by Random Forest (0.639 ± 0.120), Logistic Regression (0.604 ± 0.077) and KNN (0.591 ± 0.067). Random Forest had a slightly higher mean F1-score (0.731 ± 0.183) than Decision Tree (0.728 ± 0.219). However, Decision Tree performed best on balanced accuracy, which was used as the main evaluation metric.

Table 2: Supervised model performance across the three LOSO folds. Values are reported as mean $\pm$ standard deviation.

Model	Accuracy	Bal. accuracy	Precision	Recall	F1-score
Decision Tree	0.708 ± 0.202	0.743 ± 0.168	0.859 ± 0.116	0.658 ± 0.300	0.728 ± 0.219
Random Forest	0.673 ± 0.153	0.639 ± 0.120	0.756 ± 0.120	0.760 ± 0.302	0.731 ± 0.183
Logistic Regression	0.638 ± 0.178	0.604 ± 0.077	0.729 ± 0.085	0.704 ± 0.421	0.664 ± 0.287
KNN	0.639 ± 0.110	0.591 ± 0.067	0.731 ± 0.099	0.743 ± 0.285	0.710 ± 0.156

Mutual-information feature selection was consistent across the LOSO folds and similar across the four supervised models within each fold. Eight features were selected in all three folds, seven of which were acceleration-based, while the eighth was *gravity_y_mean*. *gravity_x_mean* and *gyro_magnitude_max* were selected in two folds. Overall, acceleration features were selected most consistently (Figure 1).

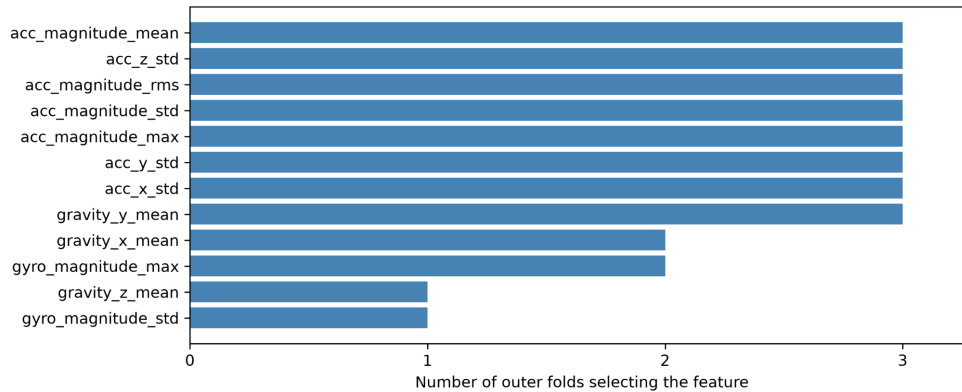


Figure 1: Decision Tree feature-selection stability across LOSO folds.

The unsupervised models formed clusters in the sensor data, but these clusters showed weak agreement with the road-condition labels (Table 3). K-Means obtained the highest silhouette score (0.400), followed by DBSCAN (0.371), GMM (0.291) and SOM (0.080). However, road ARI remained close to zero for all four models. K-Means had a road ARI of -0.029 despite its higher silhouette score, while the highest road ARI was only 0.036 for SOM.

Table 3: Performance of the unsupervised models.

Model	Clusters	Silhouette	Road ARI	Road NMI	Participant ARI	Noise (%)
SOM	36	0.080	0.036	0.144	0.097	0.0
DBSCAN	4	0.371	-0.029	0.044	0.255	9.7
K-Means	2	0.400	-0.029	0.109	0.055	0.0
GMM	2	0.291	0.035	0.094	0.137	0.0

Overall, the supervised approach was more effective for distinguishing between smooth and bumpy road conditions. Supervised learning was more suitable for the task because the aim was to assign road sections to two predefined classes. The unsupervised models identified patterns in the sensor data, but these did not correspond well to the road-condition labels. Unsupervised learning may still be useful when labelled data are unavailable. Decision Tree was selected as the final model because it achieved the highest mean balanced accuracy. It can model nonlinear relationships in the extracted numerical features, does not require feature scaling, works well with tabular data and is suitable for the relatively limited dataset with only three participants. The model is also relatively easy to interpret and requires relatively little computation during prediction.

5.3 Classification errors and failure modes

The confusion matrices show a similar error pattern for most supervised models (Figure 2). KNN, Random Forest and Logistic Regression identified bumpy windows more often than smooth windows. KNN achieved a recall of 0.78 for bumpy roads compared with 0.34 for smooth roads, Random Forest 0.80 compared with 0.40, and Logistic Regression 0.76 compared with 0.43. Decision Tree had a more balanced performance, correctly classifying 82% of smooth windows and 73% of bumpy windows.

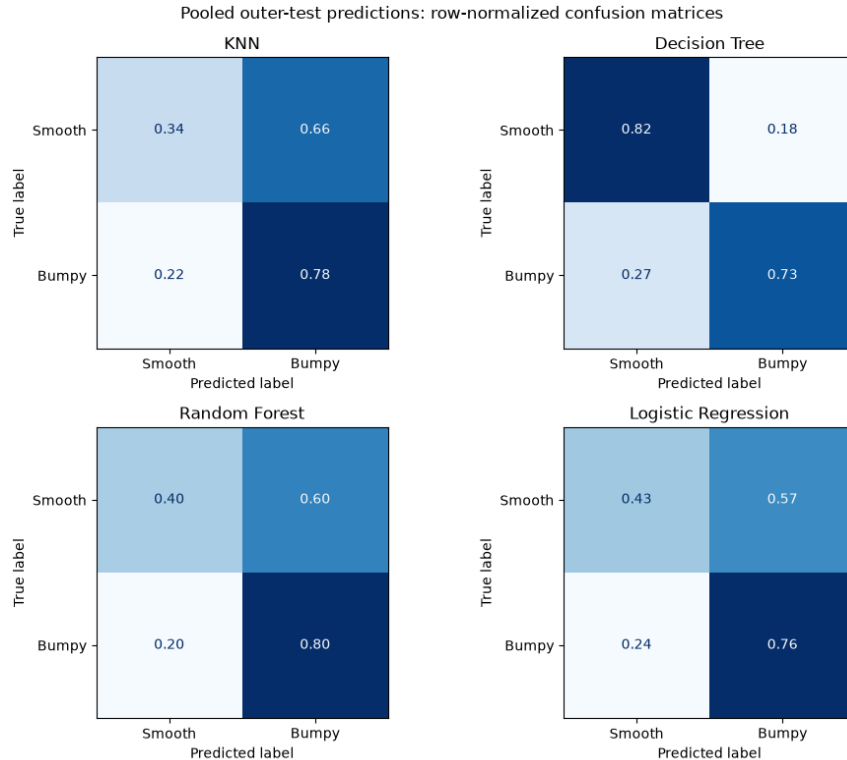


Figure 2: Confusion matrices of supervised models.

Performance also differed between participants for all four models (Figure 3). Decision Tree showed the largest variation, with balanced accuracy ranging from 0.565 to 0.899 across the three participants. KNN showed the smallest variation, with scores ranging from 0.527 to 0.661. This shows that Decision Tree achieved the highest mean balanced accuracy, but it was less consistent across cyclists.

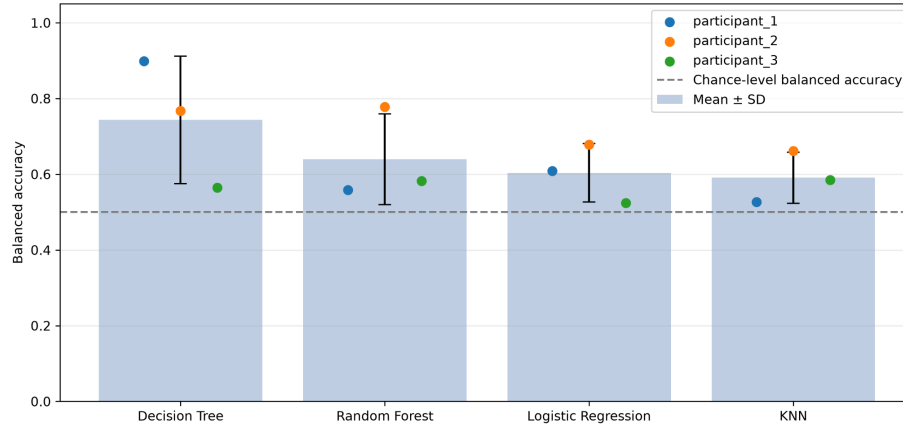


Figure 3: Table 3: Performance of the supervised models. DBSCAN metrics exclude noise, accounting for 90.3% of total windows.

The unsupervised models struggled to form clusters that represented road condition (Figure 4). K-Means and GMM both formed one cluster that was mostly bumpy, while the other cluster contained a more even mix of both road conditions. SOM produced several nodes dominated by either smooth or bumpy windows, but the road conditions were not consistently separated. Participant ARI was higher than road ARI for all four unsupervised models. The largest difference was found for DBSCAN, with a participant ARI of 0.255 compared with a road ARI of -0.029 . DBSCAN also classified 9.7% of the windows as noise.

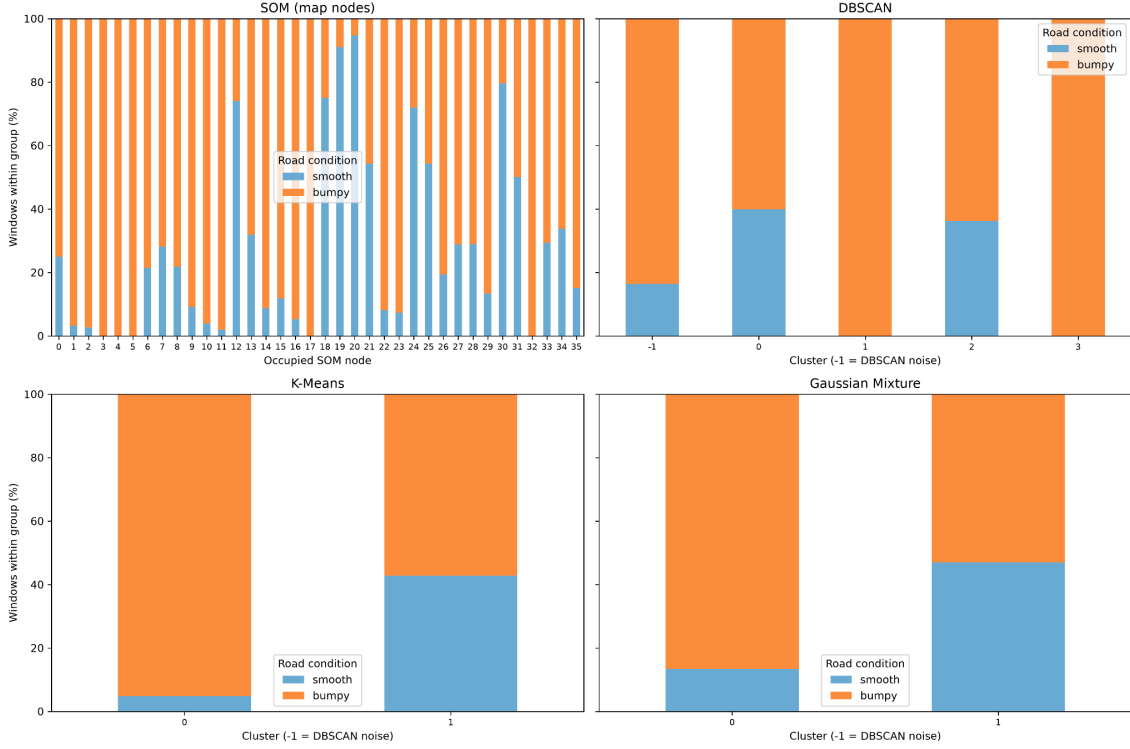


Figure 4: Road-condition distribution across unsupervised clusters.

6. Deployment Considerations & Recommendations

6.1 External dataset deployment

After selecting Decision Tree, its parameters were tuned using participant-grouped cross-validation on all labelled data. The final settings were `max_depth = 3`, `min_samples_leaf = 10`, and `class_weight = "balanced"`. The model was then fitted on all labelled data and applied to the external recording. Its decision rules used `acc_magnitude_mean`, `gravity_x_mean`, `acc_y_std`, and `acc_z_std`. The external data were not used during model training or tuning. The same preprocessing, 2-second windowing, feature extraction and feature-selection steps were applied before prediction. After preprocessing, 508 windows remained in the external recording. The model classified most of these windows as bumpy, with only short sections predicted as smooth (Figure 5). Because the external recording did not contain ground-truth road-condition labels, the accuracy of these predictions could not be evaluated.

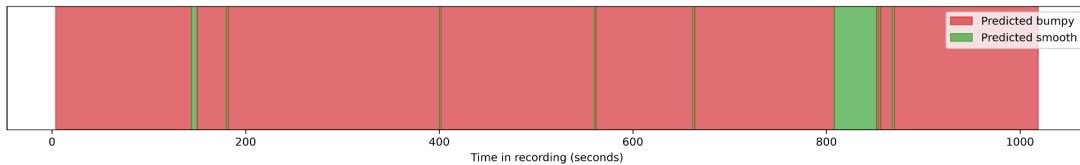


Figure 5: Predicted bicycle-lane condition on the external recording.

Deployment diagnostics also indicated a difference between the training and external data. The mean of `gravity_x_mean`, a feature used by the final Decision Tree, was 7.875 in the external dataset, compared with 0.846 for smooth and 1.863 for bumpy training windows. This difference may reflect changes in phone orientation or recording conditions rather than road quality alone. The predominance of bumpy predictions

should therefore be interpreted cautiously: without ground-truth labels, it is unclear whether it reflects actual road roughness, differences in the sensor measurements, or both.

6.2 Practical implications

Real-world deployment would require labelled training data before the Decision Tree can be used to classify new road sections. Smartphone sensors provide a relatively low-cost and scalable way to collect cycling-infrastructure data over larger areas (Ahmed et al., 2024). GPS data could be included to link smooth and bumpy predictions to specific road sections in order to identify problematic locations for further inspection (Ahmed et al., 2024). Repeated measurements from different cyclists could be combined to identify road sections that are consistently classified as bumpy. The predictions would therefore be more useful as an indication of where further inspection may be needed, rather than for making direct maintenance decisions.

6.3 Limitations and recommendations for future work

The dataset was collected by only three cyclists, and model performance varied between participants. Because each participant recorded different roads, it is difficult to know whether the differences in performance came from the cyclist or from the road itself. Future data collection should therefore include more participants and have multiple cyclists record some of the same road sections. Each complete recording also received one smooth or bumpy label, so some 2-second windows may not fully represent the assigned road condition. More detailed road-section or window-level labels would reduce this problem. The dataset was also imbalanced, with more bumpy than smooth recordings and more total recording time for the bumpy class. Collecting more smooth-road data would improve class balance. Cycling speed was not recorded, even though speed can influence acceleration measurements during cycling (Niska et al., 2024), so part of the variation in the sensor data may be related to speed rather than road condition. The phone was kept in the same pocket and orientation, but it was not fixed in place and could still move during cycling. Future work should therefore record cycling speed and secure the phone more consistently. The Decision Tree and other supervised models should be evaluated on a larger and more varied dataset across different cyclists, bicycles, phones and road types. GPS data could link predictions to locations and enable comparison across repeated rides.

References

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